

Senior Smart Contract / Blockchain Protocol Engineer

Location: Remote / Global

Employment Type: Full-time

Level: Senior

Department: Protocol & Engineering

Industry: Web3 · Fintech · Payments

Company Overview

Copia Group, LLC is building the next generation of **Web3 fintech and payments infrastructure** — essentially “**Stripe for Web3.**”

Our mission is to make blockchain-based payments and financial products as accessible, reliable, and developer-friendly as modern fintech infrastructure.

We are building technology across **smart contracts, blockchain infrastructure, wallets, payments, and financial applications**, with a focus on security, scalability, and real-world usability.

As Copia Group expands internationally, including our **Australia operation**, we are building a global engineering organization and looking for senior engineers who want meaningful ownership over foundational Web3 technology.

About the Role

We are looking for a **Senior Smart Contract / Blockchain Protocol Engineer** to design and build the blockchain layer powering our Web3 fintech and payments products.

You will work on **smart-contract architecture, protocol design, security, EVM infrastructure, transaction flows, and integrations with wallets and backend systems.**

This is a high-ownership engineering role. You will not simply implement predefined Solidity tickets — you will help determine how the protocol should be designed, secured, tested, deployed, and evolved.

What You'll Do

- Design, develop, and maintain production-grade **Solidity smart contracts**.
- Architect secure and upgradeable EVM protocols.
- Design blockchain infrastructure supporting payments, settlement, wallets, and financial products.
- Develop token, payment, treasury, vault, and financial-contract mechanisms.
- Design secure contract interfaces for frontend and backend integrations.
- Work with EVM-compatible networks and Layer 2 ecosystems.
- Analyze gas consumption and optimize contract execution.
- Develop comprehensive testing strategies for smart contracts.
- Write unit, integration, fuzz, invariant, property, and fork tests.
- Investigate complex on-chain transaction failures and protocol issues.
- Participate in security reviews, audits, threat modeling, and remediation.
- Develop deployment, upgrade, monitoring, and operational tooling.
- Collaborate closely with frontend, backend, DevOps, and product engineers.
- Evaluate new EIPs and blockchain technologies relevant to the platform.
- Contribute to protocol documentation and technical specifications.
- Mentor other blockchain engineers and establish engineering standards.

Core Responsibilities

Smart Contract Engineering

- Build secure Solidity contracts for production financial applications.
- Implement access control, permissions, pausing, emergency controls, and upgrade mechanisms.
- Design contracts with clear separation of concerns and minimal attack surface.
- Handle token accounting, settlement, fees, transfers, and withdrawals.
- Work with ERC standards and develop protocol-specific extensions where necessary.

Protocol Architecture

- Design modular blockchain systems that can evolve as the product scales.
- Evaluate architecture patterns such as:
 - Proxy / upgradeable contracts
 - Diamond architecture
 - Modular contracts
 - Vault architectures
 - Account abstraction
 - Permission systems
- Define on-chain/off-chain boundaries.

- Design reliable interaction between smart contracts, wallets, backend services, and frontend applications.

Security

Security is a first-class responsibility in this role.

You will:

- Identify attack vectors before deployment.
- Develop threat models for protocol components.
- Write security-focused tests.
- Review smart contracts for vulnerabilities.
- Work with external auditors.
- Analyze audit findings and implement remediation.
- Monitor production contracts and investigate abnormal behavior.
- Establish secure deployment and upgrade procedures.

Required Qualifications

- **5+ years of professional software engineering experience**, with significant blockchain experience.
- **3+ years of hands-on smart-contract / blockchain development.**
- Strong production experience with **Solidity and EVM**.
- Deep understanding of Ethereum transaction execution.
- Strong knowledge of smart-contract security principles.
- Experience designing and deploying production smart contracts.
- Strong understanding of ERC standards and token mechanics.
- Experience with smart-contract testing and debugging.
- Strong understanding of gas optimization.
- Experience working with blockchain RPCs, nodes, or infrastructure.
- Ability to understand and debug interactions across:
Frontend → Wallet → RPC → Smart Contract → Blockchain.
- Strong Git and CI/CD practices.
- Ability to work independently and make architectural decisions.

Web3 / Protocol Experience We Value

Experience with several of the following is highly valuable:

- Ethereum / EVM
- Solidity
- ERC-20 / ERC-721 / ERC-1155
- ERC-4626

- ERC-4337 Account Abstraction
- EIP-7702
- Upgradeable contracts
- Diamond Standard / EIP-2535
- DeFi protocols
- Stablecoins
- Crypto payments
- Tokenization
- Vaults
- Automated market makers
- Uniswap V2/V3/V4
- Layer 2 networks
- Cross-chain protocols
- Bridges
- Multisig infrastructure
- On-chain governance
- Blockchain indexing
- The Graph / Subgraphs

Testing & Security Stack

Strong experience with one or more of:

- Foundry
- Hardhat
- Forge
- Certora
- Slither
- Echidna
- Tenderly
- Mainnet / fork testing
- Fuzz testing
- Invariant testing
- Property-based testing

Experience participating in professional **smart-contract audits** is strongly preferred.

Backend & Infrastructure

The role will frequently cross into infrastructure, so experience with:

- Rust
- Go

- TypeScript / Node.js
- Python
- PostgreSQL
- Redis
- Docker
- Kubernetes
- AWS
- Blockchain nodes
- RPC infrastructure
- Indexers
- Subgraphs

is valuable, although not all are required.

Nice to Have

- Experience building **crypto payment infrastructure**.
- Experience with stablecoin settlement.
- Experience with account abstraction and smart wallets.
- Experience designing financial protocols.
- Experience with Ethereum Layer 2s.
- Experience operating production blockchain infrastructure.
- Experience with Rust or Go blockchain systems.
- Previous open-source protocol contributions.
- Experience leading protocol architecture.
- Experience working at a Web3 startup or protocol organization.
- Previous experience taking a protocol from **0** → **1**.

What Success Looks Like

First 30 Days

- Understand Copia's product architecture and protocol requirements.
- Become familiar with the existing smart-contract and blockchain infrastructure.
- Review current security, testing, and deployment processes.
- Identify architectural and security improvements.

First 90 Days

- Own significant smart-contract or protocol components.
- Deliver production-ready blockchain functionality.
- Establish strong testing and security practices.
- Work closely with application engineers on end-to-end transaction flows.

- Contribute to protocol architecture decisions.

First 6–12 Months

- Become a core technical owner of the protocol layer.
- Design major components of the Web3 payments infrastructure.
- Establish scalable smart-contract architecture.
- Help develop security and audit processes.
- Mentor engineers and contribute to protocol engineering standards.
- Help take core blockchain infrastructure from early-stage development toward production scale.

Why Join Copia Group?

- **Build “Stripe for Web3”:** Work on infrastructure designed to simplify blockchain payments and financial products.
- **Protocol ownership:** Influence fundamental architecture rather than simply implementing isolated features.
- **Security-critical engineering:** Work on systems where correctness and security genuinely matter.
- **Fintech + Web3:** Solve real financial infrastructure problems using blockchain technology.
- **Early-stage impact:** Help shape the protocol from the ground up.
- **Global expansion:** Work with an internationally oriented team as Copia Group expands, including its Australia operation.
- **Technical leadership:** Opportunity to become a foundational technical contributor as the engineering organization grows.

Ideal Candidate

The ideal candidate is a **protocol-minded senior blockchain engineer** who can take a complex financial requirement and turn it into a secure, testable, production-ready on-chain system.

You should be comfortable going deep into:

Solidity → EVM → Protocol Architecture → Security → Testing → Wallets → Transactions → Backend Infrastructure

You should also be comfortable challenging assumptions, identifying security risks, and making architectural decisions rather than simply following specifications.

If you want to help build the **blockchain protocol layer behind a Stripe-like Web3 payments platform**, this is an opportunity to have substantial technical ownership from an early stage.

